THREE EQUIVALENT FORMS OF REGULARITY FOR A SEQUENCE THAT DOES NOT GENERATE THE UNIT IDEAL

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Allowed standard prerequisites (editorial).

- Kernels of multiplication on quotients. Used at the start of the counted proof (lines 17028-17031). Turns the two-element case into a comparison of kernels.
- Regularity inherited by subsequences. Used in the induction step. Lets the induction on the length be applied to shorter sequences.

Lemma 0.1. *Let R be a ring. Let $f_1, \ldots, f_r \in R$ which do not generate the unit ideal. The following are equivalent:*

(1) *any permutation of $f_1, \ldots, f_r$ is a regular sequence,*
(2) *any subsequence of $f_1, \ldots, f_r$ (in the given order) is a regular sequence, and*
(3) *$f_1x_1, \ldots, f_rx_r$ is a regular sequence in the polynomial ring $R[x_1, \ldots, x_r]$.*

Proof. It is clear that (1) implies (2). We prove (2) implies (1) by induction on r. The case $r = 1$ is trivial. The case $r = 2$ says that if $a, b \in R$ are a regular sequence and b is a nonzerodivisor, then b, a is a regular sequence. This is clear because the kernel of $a : R/(b) \to R/(b)$ is isomorphic to the kernel of $b : R/(a) \to R/(a)$ if both a and b are nonzerodivisors. The case $r > 2$. Assume (2) holds and say we want to prove $f_{\sigma(1)}, \ldots, f_{\sigma(r)}$ is a regular sequence for some permutation σ. We already know that $f_{\sigma(1)}, \ldots, f_{\sigma(r-1)}$ is a regular sequence by induction. Hence it suffices to

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show that f_s where $s = \sigma(r)$ is a nonzerodivisor modulo $f_1, \ldots, \hat{f}_s, \ldots, f_r$. If $s = r$ we are done. If $s < r$, then note that f_s and f_r are both nonzerodivisors in the ring $R/(f_1, \ldots, \hat{f}_s, \ldots, f_{r-1})$ (by induction hypothesis again). Since we know f_s, f_r is a regular sequence in that ring we conclude by the case of sequence of length 2 that f_r, f_s is too.

Note that $R[x_1, \ldots, x_r]/(f_1 x_1, \ldots, f_i x_i)$ as an R-module is a direct sum of the modules

$$R/I_E \cdot x_1^{e_1} \ldots x_r^{e_r}$$

indexed by multi-indices $E = (e_1, \ldots, e_r)$ where I_E is the ideal generated by f_j for $1 \leq j \leq i$ with $e_j > 0$. Hence $f_{i+1} x_i$ is a nonzerodivisor on this if and only if f_{i+1} is a nonzerodivisor on R/I_E for all E. Taking E with all positive entries, we see that f_{i+1} is a nonzerodivisor on $R/(f_1, \ldots, f_i)$. Thus (3) implies (2). Conversely, if (2) holds, then any subsequence of $f_1, \ldots, f_i, f_{i+1}$ is a regular sequence in particular f_{i+1} is a nonzerodivisor on all R/I_E. In this way we see that (2) implies (3). $\square$